Ensigna Medical Center

TrialExplained

DH308-Midterm Presentation

By Angel(24B2134), Tanvi(22B0061), Madhur(24B2170), Amritesh(24B2726), Vaibhav(24B2725)

The Consequences of Flawed Design:

A Project on Clinical Trial Analysis and Implementation

The Gold Standard

Three Pillars of Integrity:

- Randomization
- Allocation Concealment
- Stratification

The Main Goal is Minimizing Bias

High cost of Failure

Case 1: The PREDIMED trial

PICOT Analysis of the PREDIMED Trial

P - Population/Patient: 7,447 individuals in Spain at high risk for cardiovascular disease (CVD). Specifically, men aged 55-80 and women aged 60-80.
I - Intervention: Two separate intervention arms, A Mediterranean diet supplemented with extra-virgin olive oil, A Mediterranean diet supplemented with mixed nuts. Both groups also received regular nutritional training.
C - Comparison/Control: A control group that was advised to follow a low-fat diet.
O - Outcome: The primary outcome was a composite of major cardiovascular events: heart attack (myocardial infarction), stroke, or death from cardiovascular causes.
T - Time: The trial was published in 2013 but was stopped early due to beneficial effects being observed.

01.

1. Failure of Randomization
The investigation revealed that randomization was improperly conducted for 21% of all participants. This is a catastrophic failure. The specific reasons were Household Clustering and Cluster Randomization Instead of Individual and Protocol Non-Adherence

02.

Failure of Stratification- The procedural randomization failure led to a de facto stratification failure

Differential Dropout (Attrition Bias): A major red flag was that more participants dropped out of the control group than the intervention groups

03.

Algorithm Design- Explicitly configured for the correct "unit" of randomization (e.g., 'individual' or 'cluster').
The randomization errors and differential dropout were only discovered years later during a post-hoc analysis
Allowing sites to use a local "randomization table" led to improper use
The investigators had to manually dig into their data to find out what went wrong

Artificially small standard errors.
Artificially narrow confidence intervals.
Artificially low p-values

Type I error, or a false positive

Case 1: The PREDIMED trial

01.

1. Failure of Randomization
The investigation revealed that randomization was improperly conducted for 21% of all participants. This is a catastrophic failure. The specific reasons were Household Clustering and Cluster Randomization Instead of Individual and Protocol Non-Adherence

02.

Failure of Stratification- The procedural randomization failure led to a de facto stratification failure
Differential Dropout (Attrition Bias): A major red flag was that more participants dropped out of the control group than the intervention groups

03.

Algorithm Design- Explicitly configured for the correct "unit" of randomization (e.g., 'individual' or 'cluster').
The randomization errors and differential dropout were only discovered years later during a post-hoc analysis
Allowing sites to use a local "randomization table" led to improper use
The investigators had to manually dig into their data to find out what went wrong

Case 1: The PREDIMED trial

pragmatic randomized controlled trial

P: Population
7,447 high cardiovascular risk adults (Spain), men 55–80 and women 60–80.

I: Intervention
Two groups received Mediterranean diets (extra-virgin olive oil or mixed nuts) + regular nutrition training.

C: Control
Low-fat diet as advised by health professionals.

O: Outcome
Composite of major cardiovascular events: myocardial infarction, stroke, or cardiovascular death.

T: Time
Published 2013; trial stopped early for benefit.

Case 1: The PREDIMED trial

Catastrophic Failure of Randomization

- 21% of participants were not randomized correctly: sites used cluster (household-based) rather than individual randomization, violating core RCT principles.
- Protocol non-adherence and the use of local randomization tables led to systematic errors.
- Instead of pure random assignment for each participant, groups were assigned based on clusters (like families), increasing the risk that characteristics unrelated to diet were equally grouped, compromising statistical independence.

Case 1: The PREDIMED trial

Stratification Failure & Attrition Bias

- The randomization error led to unintentional stratification, meaning some participant characteristics (age, risk, etc.) weren't balanced between groups.
- Significant differential dropout: control group lost more participants than intervention arms, creating attrition bias and disrupting group comparability.

Case 1: The PREDIMED trial

Algorithm and Site Practices

- Sites were allowed to run their own randomization algorithm without rigorous central oversight.
- Post-hoc data mining was required to identify errors; the randomization procedure's flaws were only discovered upon later review — not during the trial itself.
- The result was artificially small standard errors, narrow confidence intervals, and an inflated likelihood of Type I error (false positives).

RCT Randomization: Correct vs Incorrect

Aspect	Correct Randomization	Incorrect Randomization
Unit of Randomization	Individual participant assigned by chance	Groups or households assigned together (cluster)
Implementation Rigor	Strict protocol adherence, centralized oversight	Protocol deviations, local tables used
Bias Risk	Low (minimal bias)	High (systematic bias likely)
Confidence Interval Reliability	Reliable (matches statistical assumptions)	Artificially narrow, standard errors underestimated
Trial Validity	High (findings can be generalized)	Low (results questionable or misleading)

Case 2: The TROG 96.01

- P - Population/Patient: Patients with locally advanced, inoperable stage III non-small cell lung cancer.
- **I** - Intervention: Concurrent chemoradiotherapy using carboplatin.
- **C** - Comparison/Control: Concurrent chemoradiotherapy using the standard of care, cisplatin.
- **O** - Outcome: The primary outcome was overall survival. Secondary outcomes included treatment toxicity, quality of life, and tumor response rates.
- **T** - Time: The trial ran from 1996 until it was prematurely closed in 2000, with patient follow-up planned for several years.

Trans-Tasman Radiation Oncology Group (TROG)

01. **The Technical Failure (The Bug):**
The critical bug was how the software processed the date
On January 1, 2000, the software failed to correctly interpret the '00' for the year
As the date was a key input for the randomization seed, its failure caused the "random" part of the algorithm to fail

02. **The Impact of the Flaw:**
This bug completely destroyed the two most fundamental principles of a trustworthy trial, Failure of Randomization making assignment no longer based on chance and Failure of Allocation Concealment

03.
Rigorous, Adversarial Testing is Non-Negotiable
Isolate the Randomization Core from External Inputs
Implement Continuous Monitoring and Anomaly Detection
Prioritize Robustness Over Complexity

Minimization-involves stratification to ensure balance

nodal status variable

Case 2: The TROG 96.01

Data Integrity & Discrepancies

- The Denominator Mismatch: The text claims a total of 818 patients were randomized, yet the data in Table 1 sum to 820 (800 eligible + 20 ineligible), mismatching the data.
- Minimization Algorithm Pollution: Because the trial used a minimization algorithm (balancing based on previous entries), the data from these 20 ineligible patients influenced the assignment of subsequent eligible patients before they were deleted, potentially skewing the remaining cohorts.

Case 2: The TROG 96.01

Protocol Adherence & Dose Dilution

- Drug Attrition: There was a massive failure in treatment completion: 26.9% of patients in the 6-month arm and 20.1% in the 3-month arm discontinued the drug (Flutamide) early due to toxicity.
- Dose Dilution: This high dropout rate means the "6-Month Arm" effectively became a mixed cohort of variable durations, severely diluting the treatment effect and complicating the "Intention to Treat" analysis.
- Verification Bias (Missing Baselines): There was a systematic administrative failure where patients in the active treatment arms were less likely to have baseline Quality of Life data recorded compared to the control arm, introducing verification bias into the toxicity analysis

Case Study 3: The Physicians' Health Study I (PHS I) - A Near Miss with Aspirin

Randomized, double-blind, placebo-controlled

P-Population/Patient: 22,071 healthy male physicians in the United States, aged 40 to 84.
I - Intervention: 1. Low-dose aspirin (325 mg) taken every other day. 2. Beta-carotene (50 mg) taken on alternate days.
C - Comparison/Control: 1. Placebo for aspirin. 2. Placebo for beta-carotene.
O - Outcome: 1. Reduction in mortality from cardiovascular disease (for the aspirin component). 2. Decrease in the incidence of cancer (for the beta-carotene component).
T - Time: The study was initiated in 1982 and followed participants over a period of several years.

- Initial Design Flaw
- Introduction of Bias
- Compromised Groups
- The Critical Correction
- The Landmark Result

VS

- Gate the Randomization Event
- Separate Enrollment from Allocation
- Prevent Pre-emptive Allocation
- Incorporate Run-In Periods
- Create a Clear Audit Trail

Automation of patient registration and admission

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Accounting for Dropout

<u>Current methods to account for dropouts</u>

- Assume dropouts are similar to completers
- Assume last observed value is the final outcome
- Assuming baseline conditions post dropout
- Using new subjects to replace the dropouts

<u>Why do these methods fail?</u>

1. Most of the methods add bias by overestimating or underestimating the end values

- Adding new participants can break the randomisation and add bias

- Dropouts occure throughout the treatment process
- To handle dropouts often data points are extrapolated.
- This causes biaseness

<u>How colour mapping can solve this</u>

- We can attempt to predict the relative dropout factor on the basis of baseline factors and devise colours (red/yellow/green)
- Through colour mapping we then ensure each arm has a similar number of each colour

Case Study 3: The Physicians' Health Study I (PHS I) - A Near Miss with Aspirin

Randomized, double-blind, placebo-controlled

P-Population/Patient: 22,071 healthy male physicians in the United States, aged 40 to 84.
I - Intervention: 1. Low-dose aspirin (325 mg) taken every other day. 2. Beta-carotene (50 mg) taken on alternate days.
C - Comparison/Control: 1. Placebo for aspirin. 2. Placebo for beta-carotene.
O - Outcome: 1. Reduction in mortality from cardiovascular disease (for the aspirin component). 2. Decrease in the incidence of cancer (for the beta-carotene component).
T - Time: The study was initiated in 1982 and followed participants over a period of several years.

Initial Design Flaw Introduction of Bias Compromised Groups The Critical Correction The Landmark Result	VS	Gate the Randomization Event Separate Enrollment from Allocation Prevent Pre-emptive Allocation Incorporate Run-In Periods Create a Clear Audit Trail

Accounting for Blinding

Current methods to account for blinding

- By randomly allocating through computers or AI
- Using extechniques to not reveal the allocation
- Independent data monitoring boards perform unblinded safety reviews without compromising overall trial blinding.

Why do these methods fail?

- rapidly shifting allocation probabilities can create predictable patterns, repeatedly assigning the same arm and compromising blinding.

Blinding in most of the papers is not mentioned explicitly, from one of the paper Among 156 trials labeled “double blind,” only 2% clearly described who was blinded. Over half gave no details, and 26% offered no blinding information beyond the label.

How colour mapping can solve this

- We can attempt to keep reallocation randomized and unpredictable ensuring no unblinding in both patients and organisers.
- As its based on algorithms, most of it is centralised and if kept with limited access, it would be successfully able to keep blinding.

Communication channels with patients

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Our Values

01

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

02

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

03

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

95%

Success stories

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Confidentiality and security of patient Data

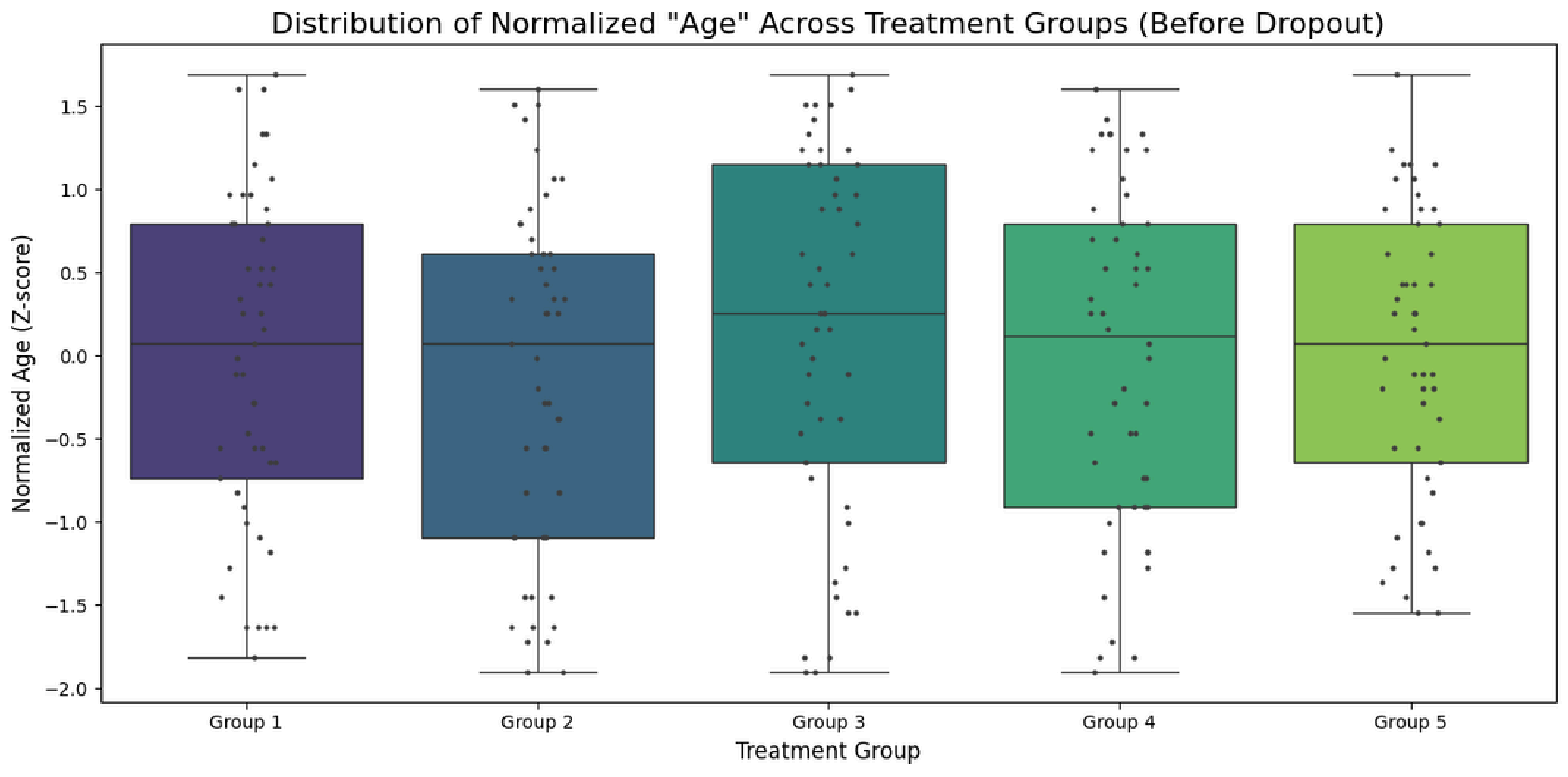
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.
Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

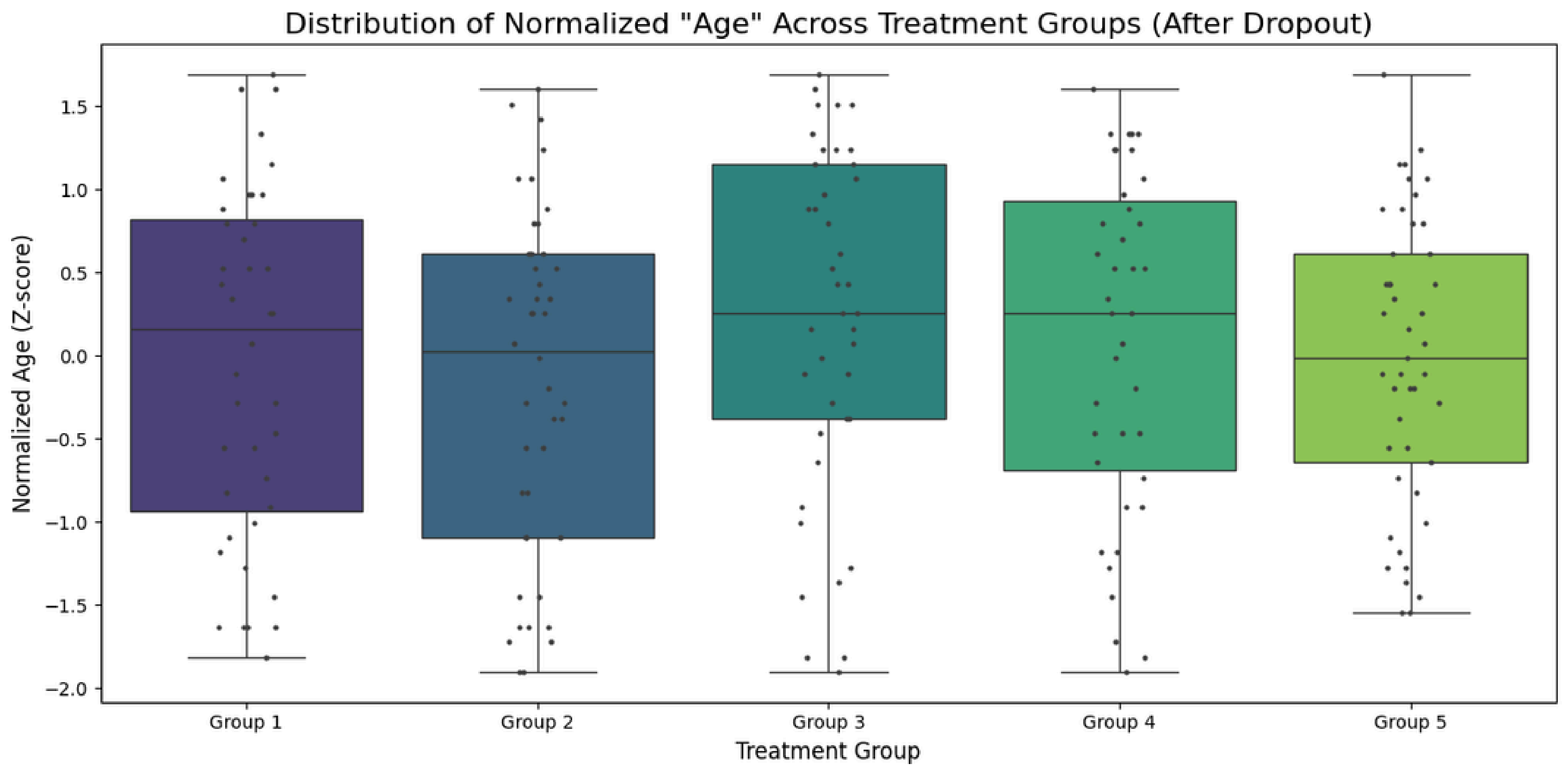
Evaluation and feedback

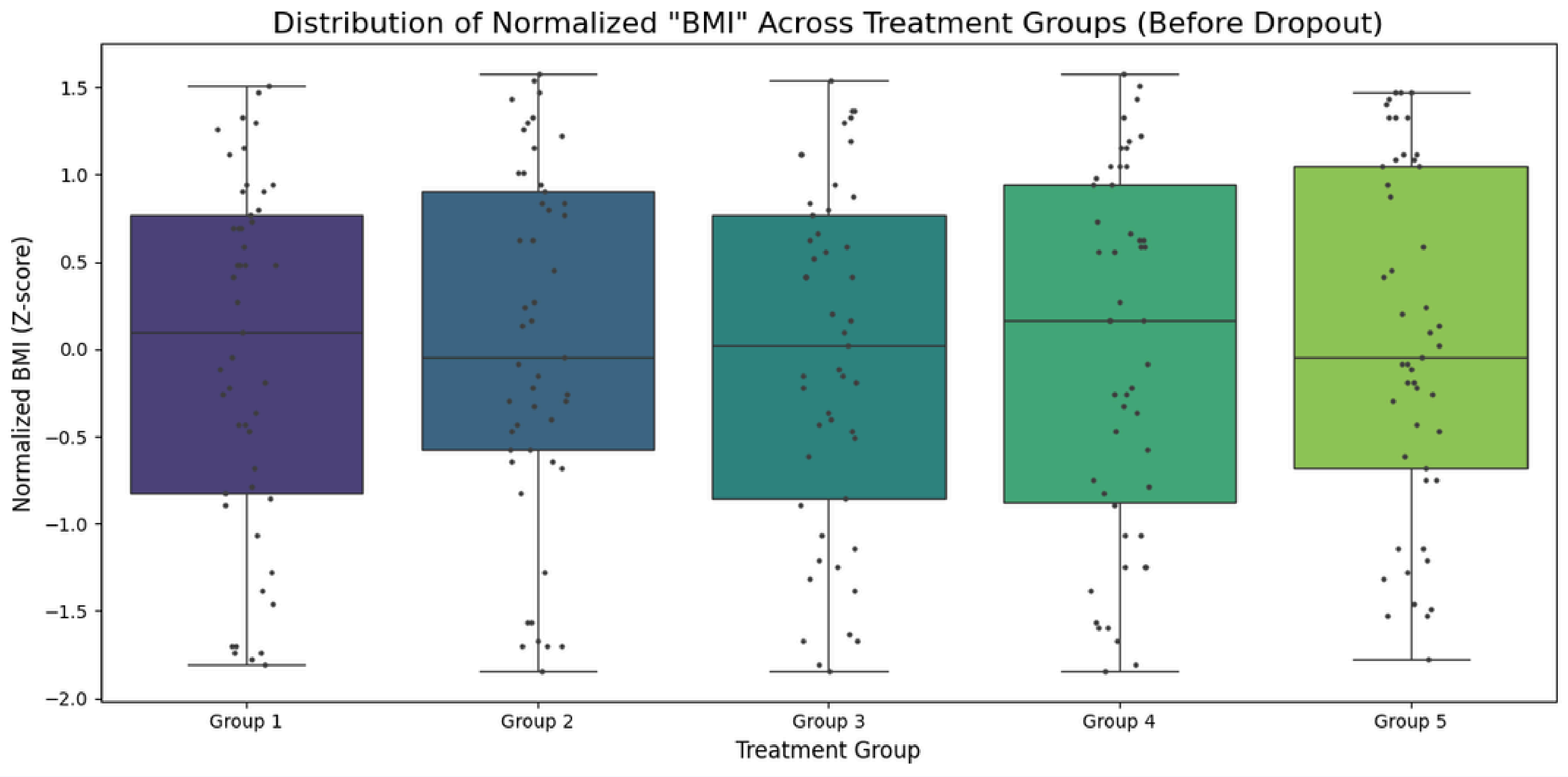
Continuously improving

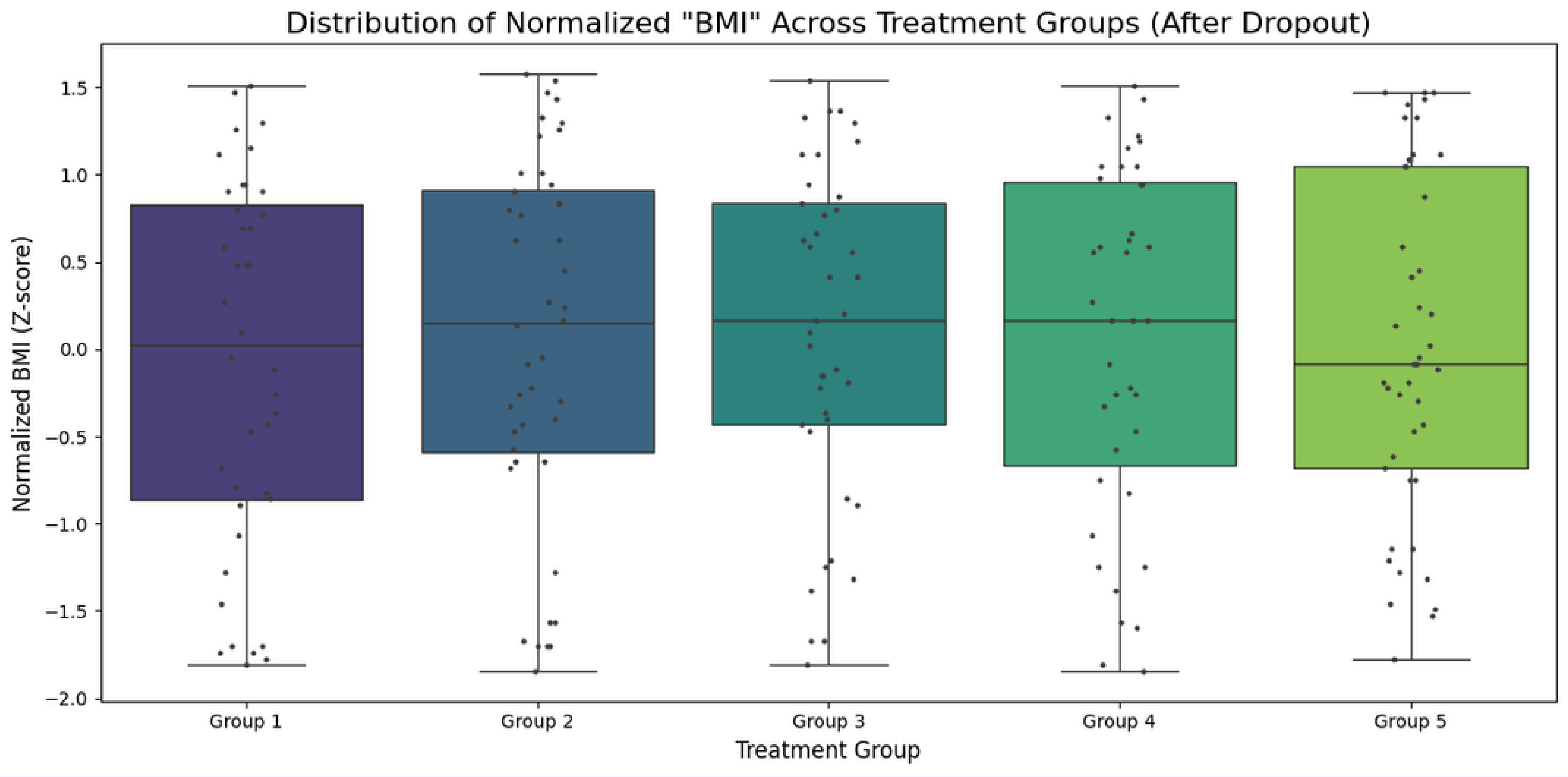
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Comparison before and after dropouts

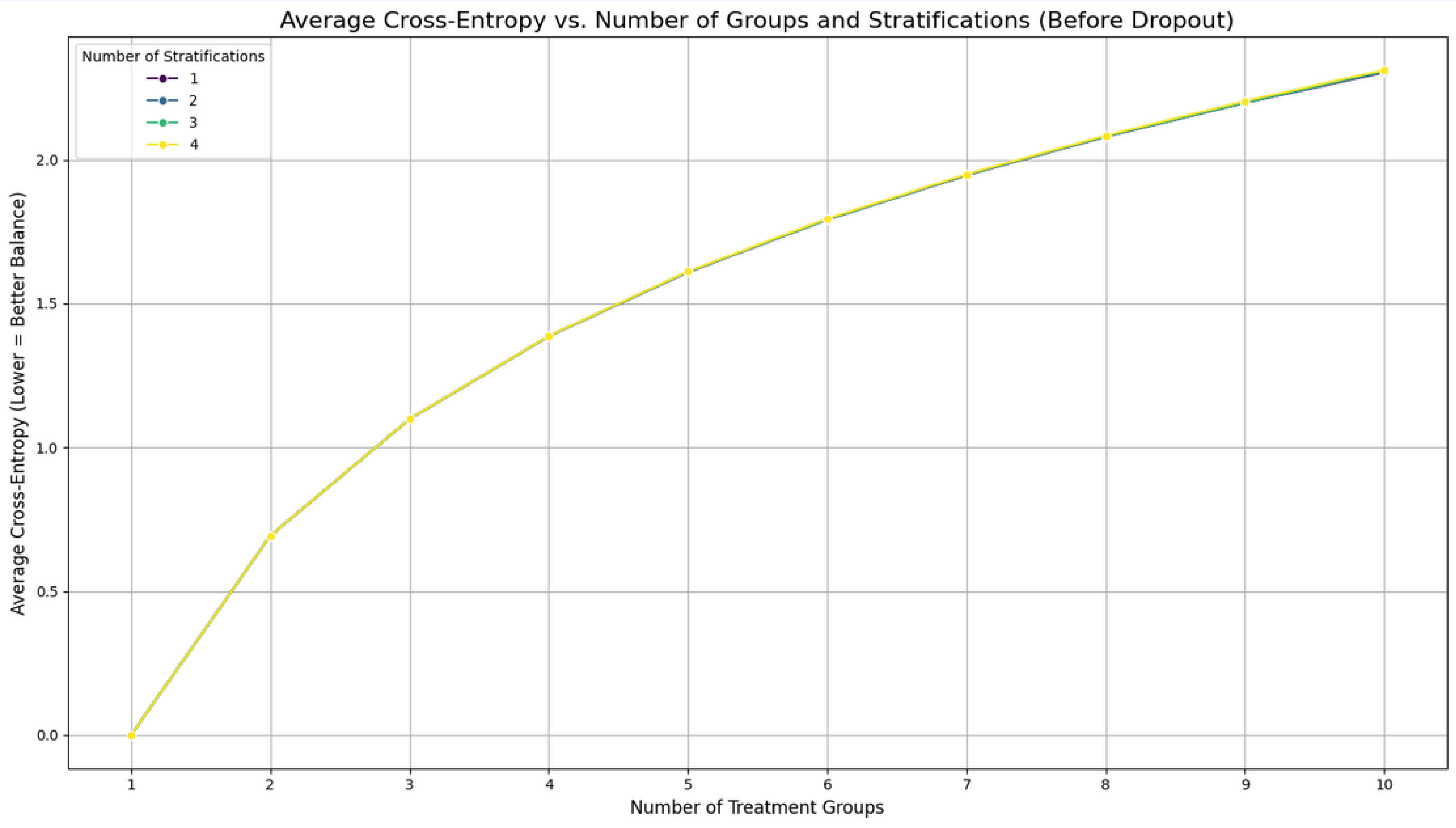


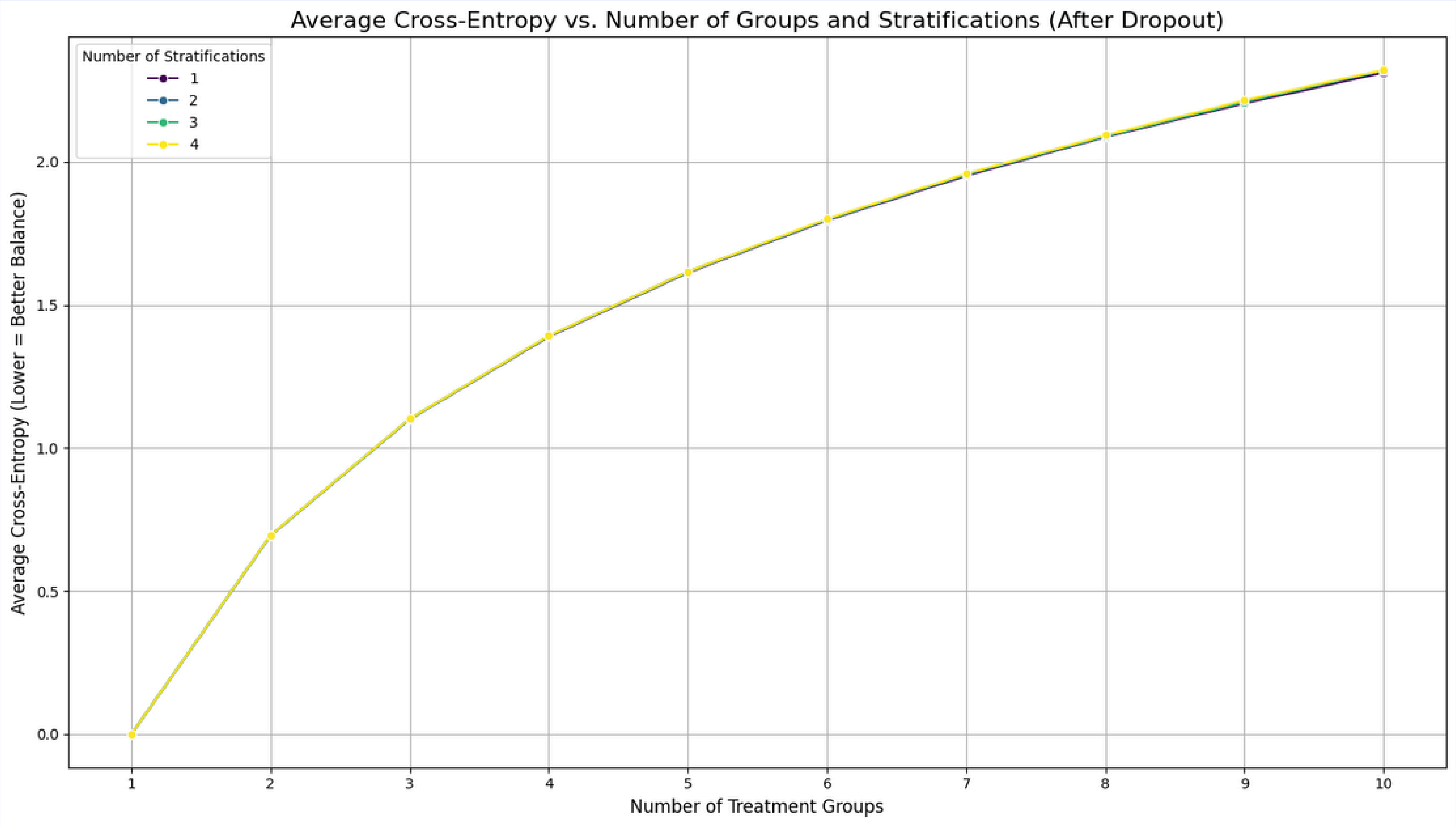






Cross Entropy metric Calculation





The metrics show that an "adversary" would not be able to break into the randomization

References

Continuously improving

1. https://absolutelymaybe.plos.org/2018/06/14/what-does-the-predimed-trial-retraction-reboot-mean-for-the-mediterranean-diet/
2. Hennekens, C. H., & Buring, J. E. (1987). The Physicians' Health Study: design and methods. Revue d'épidémiologie et de santé publique, 35(3-4), 226-235.
3. Steering Committee of the Physicians' Health Study Research Group. (1989). Final report on the aspirin component of the ongoing Physicians' Health Study. The New England Journal of Medicine, 321(3), 129-135.
4. Gebski V, Gibbs E, Simes J. The 'Y2K bug' and clinical trials: a cautionary tale. Medical Journal of Australia. 2001;175(11-12):645-647.
5. https://www.nejm.org/doi/full/10.1056/NEJM199309303291401

Ensigna Medical Center

Thank you very much!

Presented by Samira Hadid